

Electronic Trading

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Outline

- 1 Electronic markets
- 2 Simple trading algorithms
- 3 Complex trading algorithms
- 4 Case studies: when algos go wild
- 5 Conclusions

1 Electronic markets

2 Simple trading algorithms

3 Complex trading algorithms

4 Case studies: when algos go wild

5 Conclusions

Self-organised liquidity

From one to many dealers

- In the models of [Kyle and Glosten-Milgrom](#) we studied markets with:
 - ★ Several informed traders.
 - ★ Several non-informed traders.
 - ★ [Only one dealer](#).
- But in [electronic markets](#) we have a different configuration:
 - ★ Several liquidity providers: traders with Limit Orders (LOs).
 - ★ Several liquidity consumers: traders with Market Orders (MOs).
 - ★ A trader can change from provider to consumer and vice-versa.
 - ★ [No dedicated dealers](#).

How do electronic markets work without dealers?

- It is the aggregate of liquidity providers that makes the market.
 - But providers are not in obligation to place both ask and bid quotes.
- ⇒ [Self-organised liquidity](#): any trader can choose, at any instant of time, to either provide or consume liquidity.

Limit Order Books (LOB) I

Limit Orders (LO)

- When a dealer offers liquidity, she makes **quotes**:
 - ★ A **price** e.g. 14.07 EUR.
 - ★ A **quantity** e.g. 1789 shares.
 - ★ A **side** i.e. whether she buys or sells.
- In electronic markets, a quote of a liquidity provider is called a **limit order**.
- LOs can be modified or cancelled at any time.

Limit Order Books (LOB) II

Description of the LOB

- A LOB is the current liquidity offer of the market at every price level.
- ⇒ The LOB is the aggregate of all LOs.
- A LOB has 2 sides:
 - ★ **Bid side**: Prices at which the **market buys**.
 - ★ **Ask side**: Prices at which the **market sells**.
 - In electronic markets, “dealers” are anonymous and real-time data is aggregated:
 - ★ All you see are the total volumes at every price.
 - ★ But you do not know how many shares each trader inserted in the LOB.
 - For regulation, stock markets have the ID of every trader and every order.

Limit Order Books (LOB) III

Microstructure of the LOB

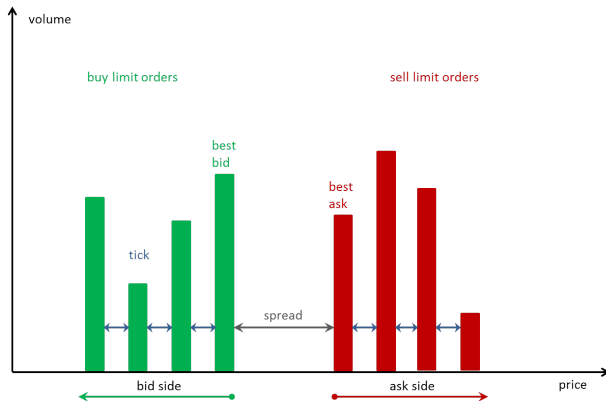
- **Priority:** Most common is price/time priority.
 - ★ **Price priority:** LOs at the best price are selected first.
 - ★ **Time priority:** Within the best price level, the oldest LOs are selected first.
- **Tick:** The smallest price increment in the LO $\Rightarrow$ **real prices are discrete**.
 - ★ In the US, the tick size for stocks is always 0.01 USD.
 - ★ In Europe, the tick size varies e.g. 0.01 EUR for prices in [50, 99.99] and 0.05 EUR for prices above 100.00 EUR.
- **Spread:** The difference between the best ask price and the best bid price.
 - ★ It is the **price of liquidity** i.e. the expected gain of a dealer.
 - ★ It is a **measure of liquidity**: the more liquid the market, the smaller the spread.
 - ★ Highly liquid markets $\Rightarrow$ spread = 1 tick.

Limit Order Books (LOB) IV

Wait a minute, prices are DISCRETE?
Do I have to de-learn all my courses and re-learn everything?



Limit Order Books (LOB) V



Sketch of a LOB.

Limit Order Books (LOB) VI

Book Viewer BZX BYX

BATS SPY Market Quality Statistics

SPDR S&P 500 ETF TR TR UNIT

Orders Accepted: 534,996 Total Volume: 4,951,439

TOP OF BOOK			LAST 10 TRADES		
	SHARES	PRICE	TIME	PRICE	SHARES
ASKS	14,100	178.7500	10:19:21	178.7100	100
	12,700	178.7400	10:19:21	178.7100	300
	20,000	178.7300	10:19:21	178.7100	200
	11,500	178.7200	10:19:21	178.7100	200
	2,200	178.7100	10:19:21	178.7100	200
BIDS	3,800	178.7000	10:19:21	178.7100	100
	21,200	178.6900	10:19:21	178.7100	100
	16,300	178.6800	10:19:21	178.7050	100
	16,200	178.6700	10:19:21	178.7050	100
	16,900	178.6600	10:19:21	178.7050	100

Last updated 10:19:24

Book Viewer BZX BYX

BATS BAC Market Quality Statistics

BANK OF AMERICA CORPORATION COM

Orders Accepted: 66,589 Total Volume: 1,567,307

TOP OF BOOK			LAST 10 TRADES		
	SHARES	PRICE	TIME	PRICE	SHARES
ASKS	32,780	16.42	10:20:48	16.37	33
	23,842	16.41	10:20:48	16.37	100
	41,255	16.40	10:20:48	16.37	800
	27,555	16.39	10:20:48	16.37	100
	30,055	16.38	10:20:47	16.37	1,000
BIDS	1,033	16.37	10:20:47	16.37	100
	20,600	16.36	10:20:47	16.37	100
	22,955	16.35	10:20:47	16.37	700
	26,755	16.34	10:20:47	16.37	100
	18,555	16.33	10:20:47	16.37	1,000

Last updated 10:20:54

Left: LOB for Spyder (ETF for S&P500). Right: LOB for Bank of America.

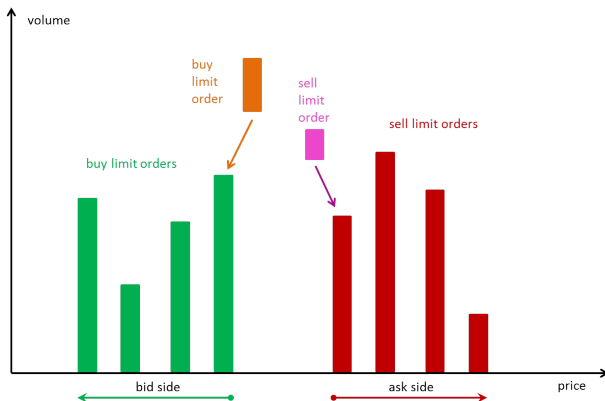
Source: <http://www.batstrading.com>

Limit Order Books (LOB) VII

Market Orders (MO)

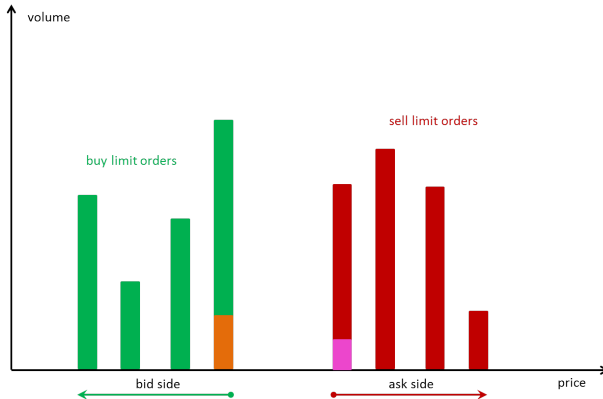
- When you send a buy market order:
 - ★ You buy at the best selling market price, i.e. the smallest available.
 - ⇒ Your order hits the ask side: **buy MO** ↔ **sell LO**.
- When you send a sell market order:
 - ★ You buy at the best selling market price, i.e. the biggest available.
 - ⇒ Your order hits the buy side: **sell MO** ↔ **buy LO**.

Inserting LOs in the LOB I



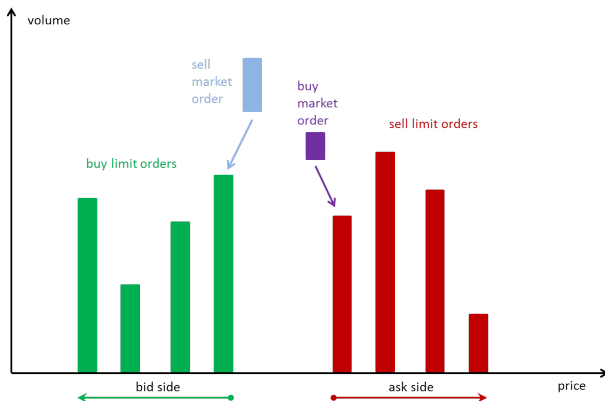
Incoming LOs: request to be inserted in the LOB.

Inserting LOs in the LOB II



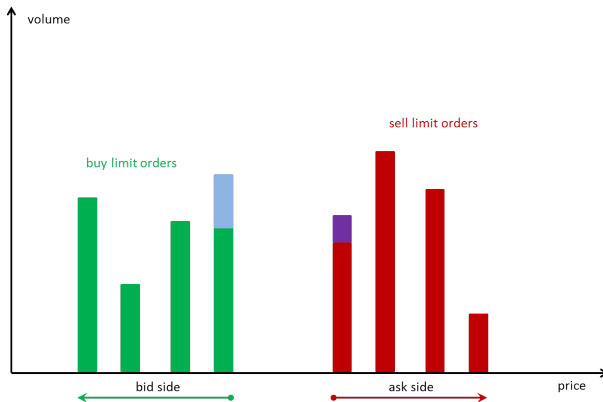
The newly inserted LOs are last in time priority.

Matching MOs vs LOs in the LOB I



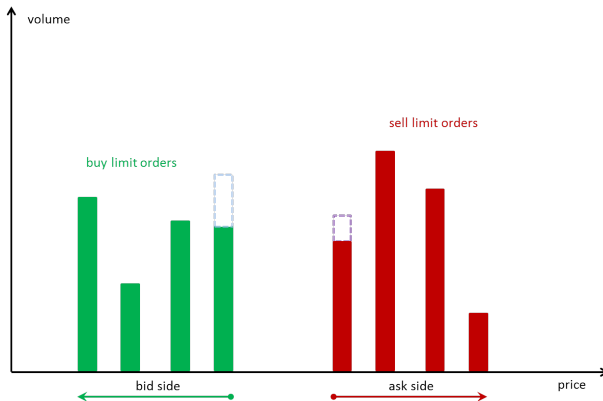
Incoming MOs: trading request at best market prices.

Matching MOs vs LOs in the LOB II

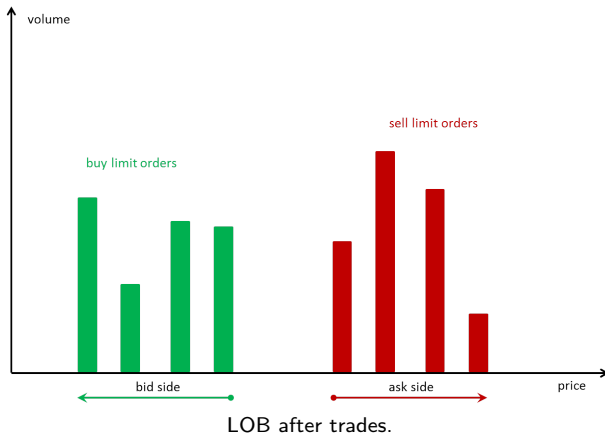


MOs are matched with LOs of highest priority price/time.

Matching MOs vs LOs in the LOB III



Matching MOs vs LOs in the LOB IV

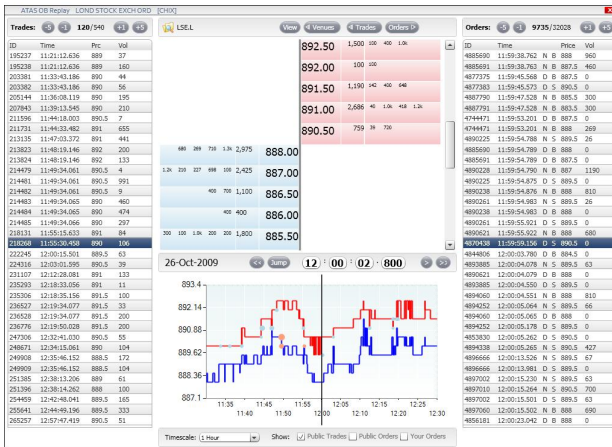


Matching MOs vs LOs in the LOB V

MOs vs LOs

- If you trade via **LO**:
 - ★ You have a **better price**: you earn the spread.
 - ★ But **delayed and uncertain execution**: your LO sits passively on the LOB, waiting to be matched with a MO.
 - ⇒ Market risk and execution risk.
- If you trade via **MO**:
 - ★ You have a **worse price**: you pay the spread.
 - ★ But **immediate and certain execution**: your MO hits the LOB and is matched vs the LOs at the best opposite price.
 - ⇒ No market risk and no execution risk.

Data flow in real time



LOB for LSE.L with tick-by-tick data at the millisecond scale.

Source: <http://www.liquidmetrix.com>

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Market-to-Limit

Market-to-Limit (MtoL)

- We send a buying LO at a price that crosses the spread e.g. the best ask.
- ⇒ We hit the ask of the LOB, as in a traditional buy MO.
- We consume all the liquidity available at the best ask.
- All non-executed shares will remain as a buy LO.
- ⇒ Our remaining shares at the old best ask become the new best bid.
- Beware: risk of incomplete execution.
- We can specify a limit price that consumes several levels of the LOB.

Fill and/or Kill

Immediate-or-Cancel (IoC) or Fill-and-Kill (FaK)

- We send a MtoL order.
- We consume all the liquidity available up to the specified price.
- All non-executed shares will be immediately cancelled.
- Beware: risk of incomplete execution.

Fill-or-Kill (FoK)

- We send a MtoL order.
 - We wait for a confirmation that our order can be fully executed.
 - ★ If the order can be fully executed $\Rightarrow$ OK.
 - ★ If the order cannot be fully executed $\Rightarrow$ we cancel the order.
- $\Rightarrow$ FoK = FaK + 100% execution guarantee.

Peg Order

Peg order

It is a LO that **always has price priority**.

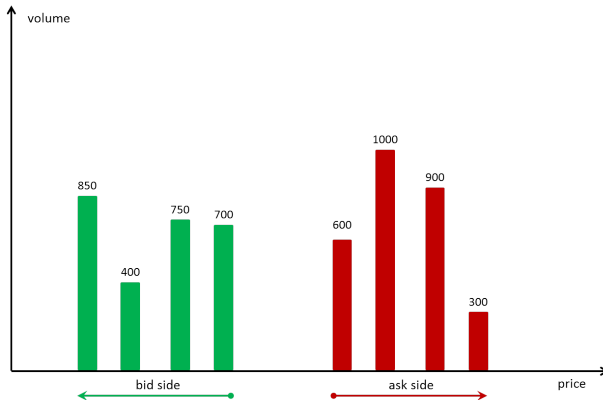
- We send a buy LO at the best bid b_0 .
- Suppose someone sends a LO and improves the best bid, say to $b_0 + \delta$:
 - ★ We have just lost the price priority.
 - ★ To recover it, we cancel our buy LO at b_0 and send a new buy LO at $b_0 + \delta$.
- We repeat the process until full execution.
- Beware: risk of incomplete execution.

Iceberg

Iceberg order

- An iceberg order of (say) 500 shares consists of two parts:
 - ★ A **visible part** of (say) 100 shares.
 - ★ A **hidden part** of (say) 400 shares.
- We place the visible part in the LOB via LO at price p_0 .
- When the visible part is executed, we send a second LO of 100 shares at p_0 .
- When the second LO is executed, we send a third LO of 100 shares at p_0 .
- We repeat the *iceberg game* until full execution:
 - ★ In our example, until the fifth LO of 100 shares at p_0 is executed.
- Beware: risk of incomplete execution.

Sketch pad for simple algos



MtoL, FaK, FoK, Peg, Iceberg.

Stop-Loss and Take-Profit I

Stop-Loss

It is the **most basic algorithm for risk management** of a position:

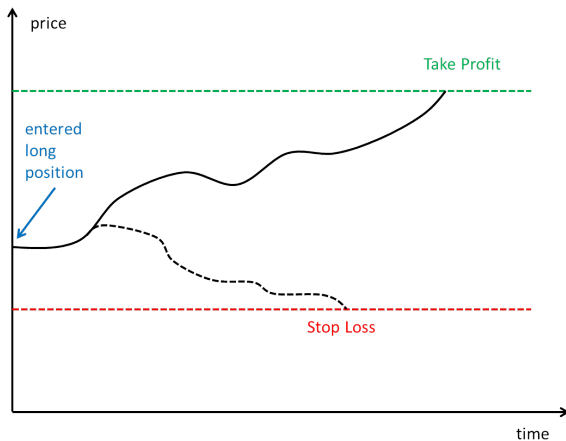
- We build up our long position.
 - We set a **maximal loss** for our portfolio.
- ⇒ Set a **minimal price or lower bound** for our portfolio.
- Portfolio value hits lower bound ⇒ we get out of the position: we sell everything.

Take-Profit

It is the **second most basic algorithm for risk management** of a position:

- We build up our long position.
 - We set a **maximal gain** for our portfolio.
- ⇒ Set a **maximal price or upper bound** for our portfolio.
- Portfolio value hits upper bound ⇒ we get out of the position.

Stop-Loss and Take-Profit II



Sketch of Stop-Loss and Take-Profit.

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Order slicing

What to do when my order is too big?

- Say we wish to clear a portfolio of 1% of total market cap of AXA:
 - ★ But 0.1–1% is the daily turnover for highly liquid stocks.
 - ⇒ Impossible to dump that quantity directly on the market.
 - There are two ways of dealing with that:
 - ★ **Block order**: we find a counterparty via a broker to clear the portfolio.
 - ★ **Slice and dice**: we cut the original order into several child orders, and send them one by one.
 - There are pros and cons for each strategy:
 - ★ With **block orders** your broker works for you, but you are revealing information: ex-ante to the broker, ex-post to the market.
 - ★ With **slice and dice** you are hiding your information from the market, but you have to do the trades yourself.
- ⇒ **Algorithmic trading and optimal execution**: what is the best way to slice and dice?

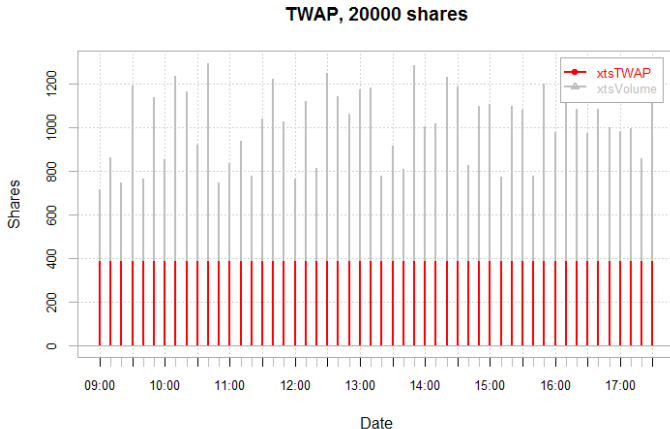
Time-Weighted Average Price (TWAP) I

Definition of the algorithm

Suppose we want to buy **20,000 shares of CActuaria** in Europe, from 09:00 to 17:30.

- We choose how many orders we want to send, say one every 10 minutes.
- ⇒ We divide the time period from 09:00 to 17:30 into 52 small periods of 10 minutes each.
- Every 10 minutes we send an order to buy $20,000/52 \sim 385$ shares.

Time-Weighted Average Price (TWAP) II



TWAP. Execution is constant throughout the whole day.

Time-Weighted Average Price (TWAP) III

Properties of the algorithm

- The TWAP is **extremely predictable**: same order (buy 385 shares) periodically sent to the market (every 10 minutes).
- To avoid being detected by the market, one can **add some randomness**:
 - ★ Random number of shares in every trade: $385 \pm \epsilon$ shares.
 - ★ Each of the 52 time periods of random time length: $10 \pm \tau$ minutes.

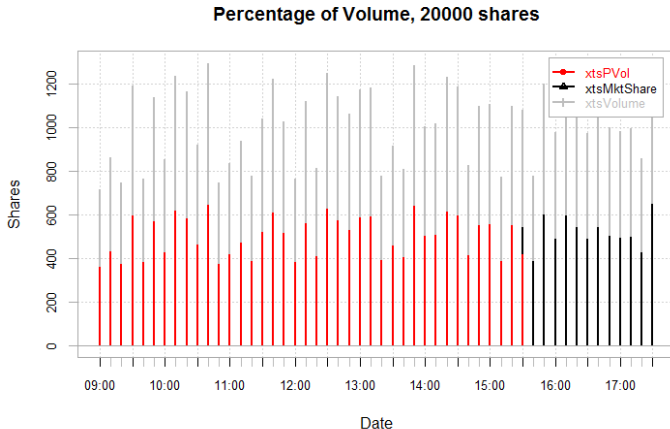
Percentage of Volume (PoV) I

Definition of the algorithm

Suppose we want to buy 20,000 shares of CActuaría in Europe, starting at 09:00.

- We compute the **intraday volume curve**:
 - ★ For each day we compute the number of traded shares every 10 minutes from 09:00 to 17:30 $\Rightarrow$ 52 periods.
 - ★ We compute the average on each period for the last month.
- We set a **market share**, i.e. the percentage of the whole market volume we will trade every 10 minutes. Let us suppose this is 50%.
- We trade 50% of the (historical) average volume every 10 minutes.
- The **final time of the algorithm** is not defined a priori: it is **part of the solution**.

Percentage of Volume (PoV) II



PoV with market share at 50%. The ending time 15:30 is part of the solution.

Percentage of Volume (PoV) III

Properties of the algorithm

- The PoV is **less predictable than the TWAP**.
- It is extremely **dependent on the intraday volume curve**:
 - ★ Robust statistical measures are needed.
- It is a **static algorithm**: the number of shares to trade is set before the execution and does not change.
- To add some **dynamic adaptation**, add price targets and flexible market shares:
 - ★ A range of market share $50 \pm \varepsilon\%$.
 - ★ A price range $p_0 \pm x$ EUR.
 - ★ A dynamic intensity: more (less) participation at low (high) prices.
- Example: suppose **CActuaria** opened the day at 100 EUR.
 - ★ If $p \in [98, 102] \Rightarrow 50\%$ participation.
 - ★ If $p < 98 \Rightarrow 75\%$ participation.
 - ★ If $p > 102 \Rightarrow 25\%$ participation.
- Beware: the **dynamic PoV has a risk of incomplete execution**.

Volume-Weighted Average Price (VWAP) I

Definition of the algorithm

Suppose we want to buy 20,000 shares of CActuaria in Europe, from 09:00 to 17:30.

- We compute the intraday volume curve every 10 minutes:

$$(V_1, V_2, \dots, V_N), \quad N = 52.$$

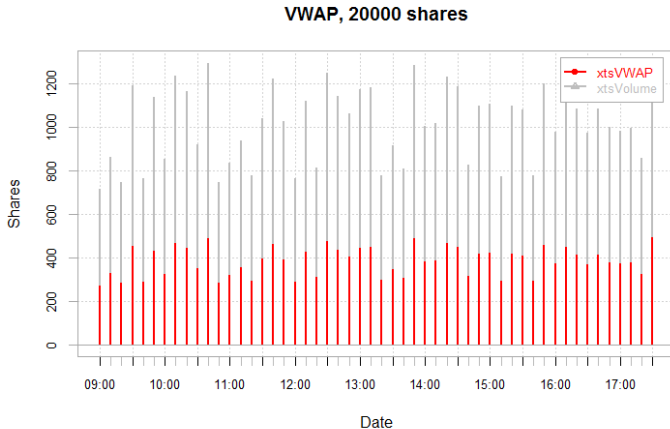
- We compute the weights at every period n :

$$w_n := \frac{V_n}{\sum_{n=1}^N V_n}, \quad n = 1, \dots, N, \quad \sum_{n=1}^N w_n = 1.$$

- We define the number of shares to buy at every period n as:

$$v_n := 20,000 \times w_n, \quad n = 1, \dots, N.$$

Volume-Weighted Average Price (VWAP) II



VWAP. Execution follows the historical (average) volume curve.

Volume-Weighted Average Price (VWAP) III

Properties of the algorithm

The VWAP shares some of the properties of the PoV:

- It is **less predictable than the TWAP**.
- It is extremely **dependent on the intraday volume curve**.
- It is a **static algorithm**.
- It can be **more dynamic with price targets**.
- The dynamic VWAP has a **risk of incomplete execution**.

However, it is **more flexible than PoV**:

- The final time N of the execution is a parameter we can play with.
- Different $N \Rightarrow$ different trading pattern.

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Flash Crash: May 6, 2010 I

Fast and Furious: Preface

- The Flash Crash lasted only **36 minutes**, but it erased 862bn USD of market capitalisation.
 - The Dow Jones lost almost 1000 points (9%) but minutes later it recovered.
 - The error was triggered by an execution algorithm *X* of a (traditional) fund. *X* activated a huge **sell order: 75,000 futures**, aprox. 4.1bn USD.
 - The asset was the E-Mini, a S&P500 future issued by the Chicago Mercantile Exchange (CME).
 - *X* was of **PoV type**, i.e. *percentage-of-volume*: it sends an order of 9% of the total traded volume of the last minute.
- ⇒ **No control on price, time or market impact.**
- Normally, an intraday execution of this size takes at least 5 hours to be filled.
 - But *X* executed it in only 19 minutes ⇒ **15 times faster.**

Flash Crash: May 6, 2010 II

Fast and Furious: Exposition

- At 2:32pm the first sell *child* order of *X* hit the market.
 - *X* consumed the market liquidity at a fast rate $\Rightarrow$ HFTs and (cross-market) arbitrageurs spotted it and started to offer more liquidity.
 - But the more liquidity the market offered to *X*, the more *X* accelerated its execution: PoV algo without upper cap.
 - The market liquidity at the bid was not enough to counter the selling pressure of *X*.
 - Between 2:41pm and 2:44pm, all HFTs capitulated: they sold aggressively 2,000 E-Mini contracts to reduce the inventories they built up by buying to *X*.
- $\Rightarrow$ HFTs not only stopped providing liquidity, they started to consume it.
- $\Rightarrow$ Chain reaction: the market liquidity at the bid disappeared, hence prices went into a free fall.

Flash Crash: May 6, 2010 III

Fast and Furious: Complication

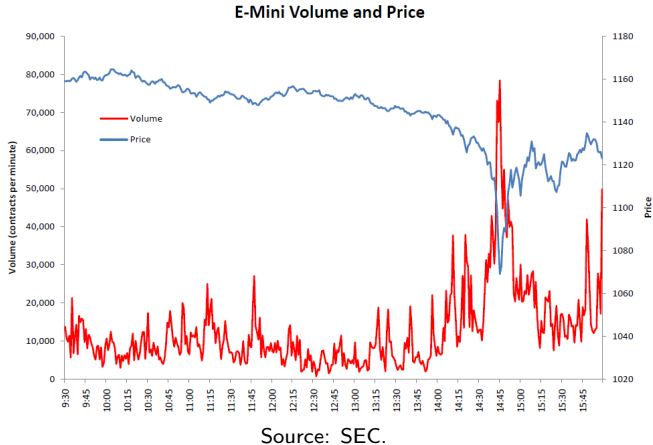
- Between 2:32pm and 2:45pm (13 minutes), with prices falling down, X sold 35,000 E-Mini contracts (1.9bn USD) out of its total of 75,000.
- During this period there were 80,000 contracts sold vs 50,000 contracts bought, i.e. a net balance of -30,000 traded contracts.
- Comparing with the former three days of trading: 15 times more sell volume and 10 times more buy volume.
- At 2:45:28pm trading in the E-Mini was suspended for 5 seconds: the CME Stop Logic Functionality triggered automatically to stop prices from further loss.

Flash Crash: May 6, 2010 IV

Fast and Furious: Resolution

- After the pause, an important flow of buy orders appeared: opportunistic and fundamental buyers.
- These buy orders not only neutralised the fall in prices, they even countered them: prices started to climb.
- But at least half of the market participants had left.
- Between 2:45pm and 3:08pm (23 minutes) more than 110,000 contracts were sold and more than 110,000 contracts were bought, but the net balance was zero.
- At 3:08pm, 36 minutes after the algo X entered the market, the intervention of opportunistic and fundamental buyers pushed prices back to their pre-crash levels.

Flash Crash: May 6, 2010 V



Knight Capital: August 1, 2012 I

Knight-mare: Preface

- Knight Capital was one of the biggest retail market-makers in the US: 10% of all trading activity.
 - A bug in the market-making software triggered wrong trading orders.
 - The error only lasted **45 minutes**, but it cost Knight **400M USD**: almost a loss of **9M USD per minute**.
 - In the end, Knight was bailed out by the private sector without any intervention of the government.
- ⇒ Damage was reputation for Knight and money of shareholders, but **zero systemic risk**.

Knight Capital: August 1, 2012 II

Knight-mare: Exposition

- On Wednesday August 1, 2012 a new market-making regulation in the NYSE started to apply.
 - Knight launched a new software to comply with the new regulation.
 - But the new software *awoke a zombie*: an old, forgotten software.
 - The *zombie* sent market orders instead of limit orders.
- ⇒ Knight consumed liquidity instead of providing it, paying the spread on each trade.
- The HFTs spotted the glitch and started to sell liquidity to Knight.
 - The *bug* took 45 minutes to be spotted and corrected, an eternity in HFT timescales.

"It was a slow, painful, 15-cents-at-a-time death by small cuts, only it was happening every fraction of a second because computers are fast."

Manoj Narang, CEO de Tradeworx Inc.

Knight Capital: August 1, 2012 III

Knight-mare: Complication

- During the *Flash Crash*, market-makers refused to provide liquidity: not clear which trades would be cancelled and which would not.
 - In the end, US stock markets and the Financial Industry Regulatory Authority cancelled 5.6M trades.
 - This upset investors, who demanded transparency and new rules.
- ⇒ New regulation to invalidate electronic trades:
- ★ For a single instrument, swings bigger than 10%.
 - ★ For 20 or more instruments, swings bigger than 30% of the whole portfolio.
- Knight asked the NYSE that each one of its 140 traded instruments would be considered individually, but the NYSE said no.
- ⇒ Only 6 out of 140 instruments had its trades invalidated.

Knight Capital: August 1, 2012 IV

Knight-mare: Resolution

- On Friday August 3, Goldman Sachs offered Knight a credit line to survive the weekend.
 - On Monday August 6, several firms paid Knight 400M USD for preferential convertible stocks (73% of Knight's market cap):
 - ★ Getco (HFT and electronic market-maker).
 - ★ Blackstone Group (hedge fund).
 - ★ Stifel Nicolaus & Co. (broker).
 - ★ TD Ameritrade Holding Corp. (broker).
 - ★ Stephens Inc. (investment bank).
 - ★ Jefferies (investment bank).
 - The *glitch* cost the original shareholders of Knight a loss of 69%.
 - But it did not cost a cent to the taxpayers.
- ⇒ In true capitalism:
- ★ High-risk $\iff$ high-reward.
 - ★ Investors are responsible: they accept profits AND losses.

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Final comments

Summary of this presentation

- We explained in detail how electronic markets work: LOBs, LOs, MOs.
- We saw simple trading algorithms: FoK, Peg, Iceberg, MtoL, Stop-Loss, Take-Profit.
- We reviewed several complex algos for optimal execution: TWAP, PoV, VWAP.
- We dissected what happened in two (in)famous case studies of algo trading going wrong: Flash Crash and Knight Capital.

References

Books

- Lawrence Harris (2002) *Trading and exchanges*. Oxford University Press.
- Barry Johnson (2010) *Algorithmic trading and DMA*. 4Myeloma Press.

Articles

- Jean-Philippe Bouchaud, Doyne Farmer, Fabrizio Lillo (2008) *How markets slowly digest changes in supply and demand*. Preprint ArXiv.
- Khalil Dayri, Mathieu Rosenbaum (2012) *Large tick assets: implicit spread and optimal tick size*. Preprint ArXiv.
- SEC (2010). *Findings regarding the market events of May 6, 2010*. Official report.

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